

RADIUS Server Setup

Part 1: FreeRadius Initialization

Step 1:

```
sudo su
sudo dnf update -y
sudo dnf check update
```

Ensures the system packages are fully updated before installing FreeRADIUS.

Running updates as root (sudo su) guarantees consistent permission handling during setup.

Step 2:

```
dnf install freeradius
```

Installs the FreeRADIUS server and all necessary dependencies.

Step 3:

```
systemctl enable radiusd.service
```

Enables the FreeRADIUS service to start automatically on system boot.

Step 4:

```
cd /etc/raddb/certs
nano ca.cnf
```

Navigates to the certificates directory.

Opens the Certificate Authority configuration file (ca.cnf) for editing.

Under [certificate authority], update relevant identification fields (like organization, email, etc.) to reflect your setup.

Step 5:

```
nano server.cnf
```

Opens the server certificate configuration file.

Modify the [server] section to match your server's identity details.

Step 6:

```
nano client.cnf
```

Opens the client certificate configuration file.

Edit the [client] section to define client certificate attributes.

Step 7:

```
make all
```

Builds and signs the CA, server, and client certificates using the configurations we just edited.

Note:

When reinitializing the server from scratch, the CA might retain old certificate data. To reset the certificate database completely:

```
cd /etc/raddb/certs
sudo rm -f index.txt*
sudo touch index.txt
sudo echo "unique subject = no" >
index.txt.attr
sudo make all
```

Removes old index files used by OpenSSL for certificate tracking.

Recreates a clean index file and allows duplicate certificate subjects.

Rebuilds all certificates to start fresh.

Permission Fix:

If running `radiusd -X` returns a "permission denied" error for `/mods-config/files/authorize`, apply the following fixes:

```
sudo chown -R radiusd:radiusd /etc/raddb
```

```
sudo find /etc/raddb -type f -exec  
chmod 640 {} \;
```

```
sudo find /etc/raddb -type d -exec  
chmod 750 {} \;
```

Assigns proper ownership of all configuration files to the `radiusd` user and group.

Ensures correct file (640) and directory (750) permissions for FreeRADIUS to read configurations securely.

Part 2: Client Connection Setup

Step 1:

```
nano clients.conf
```

Opens the client configuration file to define trusted RADIUS clients.

Step 2:

Example test client entry to add in `clients.conf` file.

```
client justin {  
    ipaddr = 34567u76\  
    secret = whatever  
}
```

Defines a test client with its IP address and shared secret. This secret must match the one used by the client when sending authentication requests.

Step 3:

```
nano /etc/raddb/mods-config/files/authorize
```

Opens the user authorization file to define local test accounts for authentication.

Step 4:

Example test entries to add in authorize file.

```
    justin      Cleartext-Password :=  
test  
    guru        Cleartext-Password := test
```

Creates two test users (justin and guru) with plaintext passwords for simple authentication testing.

Step 5:

Now run the RADIUS server

```
radiusd -X
```

Note:

When having issues with starting radiusd -X due to port being used:

```
sudo lsof -i :1812 1813
```

Lists processes currently using the RADIUS ports (1812 for authentication, 1813 for accounting).

Useful for identifying and stopping any running instances that block radiusd -X from starting.